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Education

National Chiao Tung University
B.S., Applied Mathematics (**Top 1%**)

UC Berkeley
M.A., Applied Statistics

SIAM in Taiwan

Enzyme condition optimization

- **Best poster paper** at TwSIAM

NTU Hospital

Healthcare Project

- **Save 1,600 people per year**

Taiwan Semiconductor

Manufacturing Company (TSMC)

- **Saving \$100,000** per week

Experience

Research Highlights

1. Spectral identifiability and stability.
2. Fisher curvature as a stability measure.
3. Geometry, statistics, and optimization in learning systems.

Research Interests

- Statistical Learning Theory & Identifiability
- Optimization Stability & Dynamics
- Reliable & Interpretable learning

Spectral Identifiability: Conditioning, Stability, and Sensitivity

Stability is governed by geometry (conditioning), not the optimization algorithm.

Framing

Problem setup:

- Retrained networks are accurate but inconsistent.
- Explanations drift is a geometric identifiability issue.

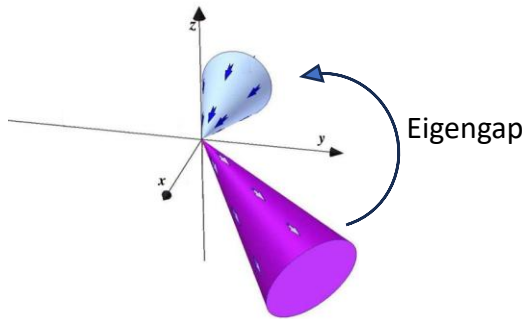
Core Message:

- Stability is geometric, not algorithmic
- If eigengap > perturbation, solution is well-conditioned.

Implication:

- Identifiability $\Rightarrow$ reproducible explanation.
- Eigenspace geometry = credibility measure.

Task directions are separated from noise by the spectral gap.



Definition:

- Representation X = output of a fixed network layer
- Defines $\Gamma = XX^T$ as the representation geometry

Spectral Identifiability Principle (SIP)

Key idea:

- If fisher gap defines then solutions stays stable.

With moderate conditions,

$$gap(\Gamma) = \lambda_k - \lambda_{k+1} > 0, \|h(X)\| \leq B,$$

$(h(X) : \text{output of a fixed representation layer, } \Gamma = E[h(X)h(X)^T])$

The following holds,

1. Subspace concentration (Davis-Kahan)

$$\sin \Theta(\hat{U}, U) \leq \frac{\Delta}{gap(\Gamma)}$$

(Δ : Finite-sample perturbation)

2. Risk bound

$$\Pr(\hat{f}(X) \neq f(X)) \leq C \cdot B^\kappa \cdot \min(1, (\frac{\Delta}{gap(\Gamma)})^\kappa)$$

(κ : sensitivity of prediction error)

3. Sample complexity

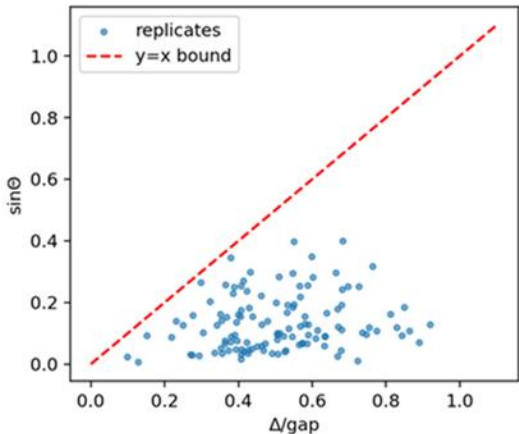
$$\text{if } n \geq \frac{\log(d)}{gap(\Gamma)^{-2}}, \text{ error decays with } \tilde{O}(n^{-\kappa/2})$$

Consequences of Spectral Identifiability

A single spectral ratio governs conditioning, error sensitivity, and data requirements.

Theory	Interpretation
Subspace concentration	Subspace error is controlled by $\Delta/gap(\Gamma)$
Risk bound	Classification error decays as $(\Delta/gap(\Gamma))^\kappa$
Sample complexity	To let Δ small, we need $n \geq \log(d) / gap(\Gamma)^{-2}$

Observed subspace deviation follows the Δ/gap scaling.



From Identifiability to Stability: A Spectral Threshold Perspective

- Fisher curvature defines when learning remains dynamically stable under noise.

Framing

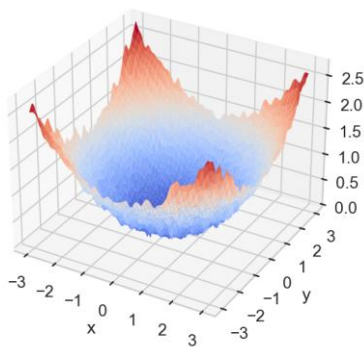
Problem: Dynamic instability

- Training loss fluctuates under noise
- Local minima drift across runs
- Static identifiability $\neq$ dynamic stability

Insight: Fisher curvature guides local optimization

- $\lambda_{min}(\Gamma)$ acts as curvature floor
- Above floor $\rightarrow$ linear convergence
- Below floor $\rightarrow$ PL fails

Below the threshold, the loss landscape flattens and local minima become indistinguishable.



Definition:

- X is the centered score matrix
- Fisher matrix $\Gamma = E[XX^T]$ defines the curvature.

Spectral Thresholds for Stability

With moderate conditions and define

$$\Lambda^* = C\sigma_{eff} \sqrt{\frac{d + \log(1/\delta)}{n}},$$

(n : sample size, δ : failure probability, d : effective dimension, σ_{eff} : noise level)

the following holds

1. Above-threshold

if $\lambda_{min} \geq 2\Lambda^*$, learning is Polyak–Łojasiewicz,
 $\|\nabla L\|^2 \geq 2\mu(L - L^*)$.

(λ_{min} : smallest fisher eigenvalue, L : training loss function)

2. Below-threshold

if $\lambda_{min} \leq \frac{1}{2}\Lambda^*$, then local become indistinguishable.

3. Constructive Fisher floor

With regularizer, $R_\tau(\theta) = \max_{\|u\|=1} (\tau - u^T \hat{\Gamma}(\theta)u)^2_+$,

stability can be enforced to lift the Fisher spectrum.

(λ_{min} : target curvature floor, $\hat{\Gamma}$: empirical fisher estimate)

Consequences of Spectral Thresholds

All effects are governed by the same spectral threshold, Λ^* .

Theory	Interpretation
Above threshold	If $\lambda_{min} \geq 2\Lambda^*$, training converges exponentially.
Below threshold	If $\lambda_{min} \geq \frac{1}{2}\Lambda^*$, noise dominates.
Constructive Fisher floor	A regularizer penalizes directions below threshold.

Accuracy stabilizes once the Fisher curvature exceeds the spectral threshold.

